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Where I Becomes Infinite

Top-P (Nucleus Sampling)

Module 3.1: Working with LLM APIs

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Session Overview

Prerequisites

- Probability distributions over tokens
- Softmax function and logits
- Temperature parameter and its effect
- Cumulative probability and sorting

What You Will Learn

- What nucleus sampling is and why it was invented
- The step-by-step algorithm for top-p filtering
- How top-p adapts to distribution shape dynamically
- Practical top-p values and their effects
- Comparison with top-k and temperature

The Problem Top-P Was Designed to Solve

Why Greedy and Fixed-Window Sampling Fall Short

Greedy decoding always picks the single most likely token, leading to **repetitive, generic** text. Uniform sampling over all tokens produces **incoherent** text by including highly improbable tokens. Top-k sampling (covered next) uses a fixed window size, but a fixed window of 40 tokens may be too wide when the model is confident, and too narrow when uncertainty is high.

The Core Insight of Nucleus Sampling

- The model's uncertainty varies step by step
- When the model is confident: 3–5 tokens hold nearly all probability
- When the model is uncertain: 50–100 tokens may each have 1% probability
- A **fixed-size window** (top-k) cannot adapt to this variability
- Top-p uses a **probability mass threshold** instead of a count, automatically adapting the window size to the distribution's shape

The Nucleus Sampling Algorithm

Top-P Filtering — Step by Step

1. **Compute probabilities:** Apply temperature-scaled softmax to get $p_i(T)$ over the full vocabulary ($\sim$100K tokens)
2. **Sort descending:** Order tokens from highest to lowest probability
3. **Find the nucleus:** Starting from the top, accumulate probability mass until the cumulative sum first exceeds or equals threshold p
4. **Define the nucleus set $\mathcal{N}_p$:** The minimal set of tokens whose cumulative probability $\geq p$
5. **Re-normalise:** Redistribute probability mass uniformly over $\mathcal{N}_p$, setting all other token probabilities to zero
6. **Sample:** Draw one token from the re-normalised nucleus distribution

Nucleus Sampling --- Visual Illustration

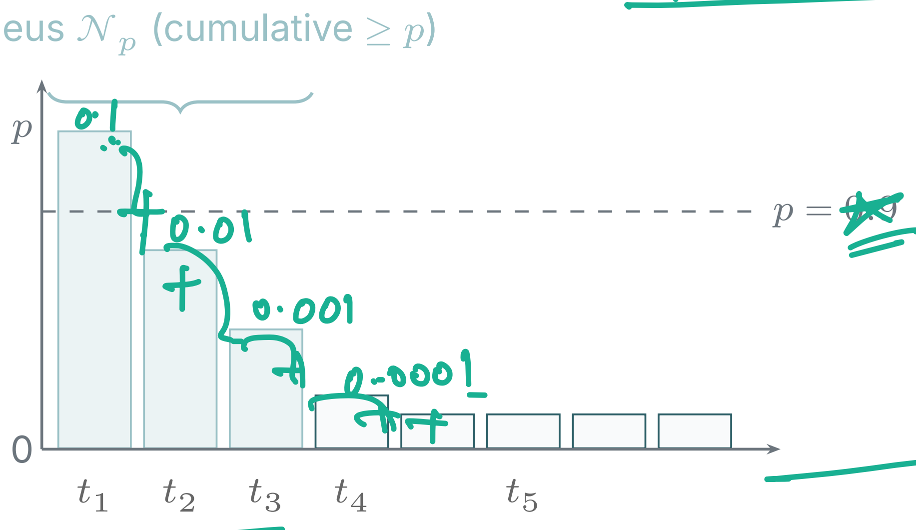
$\boxed{top\_p = 0.11105}$

$CP_1 = 0.1$

$CP_2 = 0.11$

$CP_3 = 0.111$

$CP_4 = 0.1111$



0.111

$CP_3 < top\_p$

$CP_4 > top\_p$

Tokens t_1, t_2, t_3 form the nucleus when $p = 0.9$. The remaining tokens are excluded and their probability redistributed.

Adaptive Window: The Key Advantage

Confident Prediction ($p = 0.9$)

Distribution: $p_1 = 0.88$, $p_2 = 0.07$, $p_3 = 0.03$, ...

Cumulative sum ≥ 0.9 after **2 tokens**.

The nucleus contains just 2 tokens.

Effect: Highly constrained, near-deterministic. Appropriate because the model "knows" what comes next.

Uncertain Prediction ($p = 0.9$)

Distribution: $p_1 = 0.02$, $p_2 = 0.018$, ..., many equal tokens

Cumulative sum ≥ 0.9 after **48 tokens**.

The nucleus contains 48 tokens.

Effect: Wide sampling. Appropriate because many continuations are equally plausible.

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Top-P Values --- Practical Reference

Top-P	Label	Nucleus Size (typical)	Use Case
0.1	Very tight	1–3 tokens	Near-greedy; structured extraction
0.5	Moderate	5–20 tokens	Focused, coherent generation
0.75	Balanced	15–50 tokens	General assistant interactions
0.9	Default	20–100 tokens	Recommended default for most tasks
0.95	Creative	50–200 tokens	Story writing, brainstorming
1.0	No filtering	Full vocabulary	Effectively disables top-p

The nucleus size is dynamic and varies per step even at a fixed p. Most LLM providers default to `top_p=1.0` (disabled) when temperature is set.

Nucleus Sampling vs. Top-K Sampling

Property	Top-P (Nucleus)	Top-K
Window definition	Probability mass threshold	Fixed count of tokens
Window size	Adapts to distribution shape	Always exactly k tokens
When model is confident	Small window (2–5 tokens)	Always k tokens (may be too wide)
When model is uncertain	Large window (50–100 tokens)	Always k tokens (may be too narrow)
Parameter intuition	"Include enough tokens to cover p% of mass"	"Always consider top k options"
Recommended default	`top_p = 0.9`	`top_k = 40--50`

Top-p adapts to context; top-k applies a rigid constraint. Both are applied *after* temperature scaling.

Parameter Interaction Best Practices

How to Set Temperature and Top-P Together

- **OpenAI recommendation:** Adjust `temperature` or `top_p`, not both. Each achieves similar diversity control; combining them makes behavior harder to predict and tune
- **Temperature first:** Start with temperature to get the desired creativity level; use top-p as a ceiling to prevent catastrophic tokens
- **Top-p as safety guard:** Even with temperature=1.0, setting `top_p=0.95` prevents the bottom 5% of the vocabulary (highly unlikely, often incoherent tokens) from ever being sampled
- **Anthropic's default:** Claude API defaults temperature to 1.0 with top-p as the primary diversity control
- **Evaluation tip:** Run the same prompt 10 times; measure output variance to validate your parameter combination empirically

GenAI Application: Default Sampling in GPT-4o

How OpenAI Configures Top-P for ChatGPT

- **ChatGPT interface:** Uses temperature=1.0 with top-p=1.0 by default — relying on the raw RLHF-trained distribution to be appropriately calibrated for helpful, harmless responses
- **API default:** `top_p=1` when not set, effectively disabling nucleus filtering and relying on temperature alone
- **GPT-4 code interpreter:** Internally uses low temperature with tight top-p for deterministic code execution tasks
- **DALL-E 3 captions:** Uses high temperature + top-p≈0.95 when generating diverse creative descriptions of images
- **Whisper transcription:** Temperature=0 for deterministic audio-to-text; top-p is not applicable for transcription tasks

Production defaults are carefully calibrated by providers through extensive RLHF tuning.

GenAI Application: Top-P in Anthropic Claude

Anthropic's Nucleus Sampling Implementation

- **Claude API:** Exposes `top_p` as a first-class parameter alongside temperature; API documentation recommends using one or the other for most use cases
- **Claude for enterprise:** System integrators tune top-p per endpoint — `top_p=0.3` for policy Q&A, `top_p=0.95` for open-ended conversational assistants
- **Constitutional AI interaction:** High top-p combined with Claude's constitutional training maintains diversity without sacrificing safety alignment — the RLHF-trained distribution already down-weights harmful tokens
- **Long-form writing:** Claude's 200K context window combined with top-p ≈ 0.9 produces coherent, non-repetitive long documents where low top-p would cause the model to reuse the same phrases

Claude's documentation explicitly warns against setting both top-p and temperature simultaneously.

GenAI Application: Open-Source LLM Serving

vLLM Serving Defaults

- vLLM (Meta/community) serves Llama 3, Mistral, Qwen
- Default: temperature=0.7, top-p=1.0, top-k=-1 (disabled)
- Ollama local serving: top-p=0.9 as default across all models — practical default for most local workloads
- LM Studio: Exposes all sampling parameters through UI sliders, making nucleus sampling directly observable

Open Model Fine-Tuning Consideration

- Fine-tuned models may have *different* optimal top-p compared to base models due to distribution shift
- Instruction-tuned models (Llama-Instruct) are calibrated for lower temperature; may need tighter top-p
- Evaluate sampling parameters on a held-out validation set when deploying a newly fine-tuned model
- Quantised models (GGUF/GPTQ) may need slightly higher top-p to compensate for quantisation noise

GenAI Application: Structured JSON Output Generation

Top-P for Schema-Compliant Outputs

- **Function calling (OpenAI, Anthropic):** When generating structured tool parameters, providers use internal logit bias mechanisms alongside low top-p to enforce JSON schema compliance
- **Outlines / Guidance:** Open-source constrained generation frameworks mask illegal tokens at each step. Combined with top-p, this ensures diversity within the schema (different valid values) while eliminating schema-violating tokens
- **Named Entity Recognition (NER):** LLMs fine-tuned for NER use top-p=0.1–0.3 with specific label vocabularies to produce clean, consistent entity labels without hallucinating new categories
- **SQL generation:** Text-to-SQL systems use low top-p (≤ 0.5) combined with temperature=0.1 to ensure syntactically valid SQL with minimal random token intrusions

Session Summary

Key Takeaways

1. **Nucleus sampling** (top-p) restricts sampling to the smallest set of tokens whose cumulative probability $\geq p$
2. Unlike top-k, the nucleus is **adaptive** — it shrinks when the model is confident and expands when uncertain
3. Top-p is applied *after* temperature scaling and *before* the final token is sampled
4. **Recommended default:** `top_p=0.9` for most production applications; lower for structured tasks
5. Do not set both temperature and top-p unless you understand their combined effect on distribution shape
6. Top-p and top-k are complementary: top-p adapts to shape; top-k enforces a hard count ceiling

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